Research Whitepaper

Seventeen Trading Ideas, Tested Honestly on Free Data

A Pre-Registered Negative-Results Study of Short-Term US Stock Strategies

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The Sift project · github.com/Malekhalifa/sift

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Abstract

Trading strategies spread online because they are easy to describe and easy to believe. This paper asks whether any of them survive a fair test, starting from $0 and using only free data. It describes **Sift**, a research pipeline that downloads market data, turns each trading idea into exact rules, simulates the trades with realistic costs, and judges the results with checks designed to catch luck: pre-registration of every protocol before it runs, walk-forward testing, an untouched holdout, a count of every configuration tried with a Deflated Sharpe Ratio, a comparison against simply holding the market, and replication on unseen stocks. Seventeen ideas were tested on up to 410 US stocks from 2016 to 2024: opening-range breakouts and fades, RSI reversion, trend pullbacks, earnings drift, insider buying, two indicator bots popular in online videos, volume-conditioned rules, presidential social-media posts and Pentagon contract awards, and, on new free data added later, 8-K filings, 52-week highs, short-sale volume, share issuance and the wording of 10-K risk factors. **None met all of its pre-registered criteria.** The ideas that looked strongest were either the market's own rise or luck that did not repeat on new stocks. The contribution is a method and a set of lessons about how backtests flatter themselves, not a strategy.

Disclaimer. A personal research project. It is not investment advice and recommends no security or strategy; no real money was traded. The code, analysis and drafting were done with an AI coding assistant; every number comes from results recorded in the project repository, whose commit history records when each protocol was written.

Contents

Executive summary **2**

1 Introduction **3**
1.1 Motivation . . . 3
1.2 How Sift works . . . 3
1.3 Constraints fixed at the start . . . 4

2 The seventeen ideas **5**
2.1 Short-term price patterns (minutes to hours, 5-minute bars) . . . 5
2.2 Daily swing trades (days to weeks, daily bars) . . . 5
2.3 Events and news (daily bars) . . . 5
2.4 Indicator bots popular in online videos (5-minute bars) . . . 6
2.5 Volume (daily bars) . . . 6
2.6 Five more ideas on new free data (milestones M14 to M18) . . . 6

3 Methodology **7**
3.1 Reading a result . . . 7
3.2 How a trade is simulated . . . 8
3.3 How walk-forward works . . . 8
3.4 What the Deflated Sharpe check asks . . . 8

4 Results **9**
4.1 Reading the table . . . 10
4.2 Which safeguard stopped each idea . . . 11
4.3 The two event studies (M12) . . . 11
4.4 Five more tests (M14 to M18) . . . 12

5 Discussion: what kept recurring **13**
5.1 A closer look: the near miss . . . 15

6 A side question: does a trend filter make holding the market safer? **15**

7 Limitations **15**

8 Conclusion **16**

9 Recommendations **16**

10 Reproducibility **17**

A Glossary **18**

B Index of protocols and results **19**

Disclosures **20**

References **20**

Executive summary

Sift was built to answer a practical question: can a retail trader with no capital and only free data find a short-term edge in US stocks, and prove it before risking money? Seventeen ideas were each turned into exact rules, written down before testing, and judged out of sample by fixed, pre-registered criteria. The 2025 to 2026 data was locked away for a final check that no idea earned.

17	0	304	410	9
ideas tested	passed every check	configurations counted	US stocks at most	years of data

Key findings

- **No idea passed.** The two closest, earnings drift and a high-volume rule, each passed five checks and then failed the decisive ones.
- **Apparent edges were usually the market.** Long-only rules looked positive because stocks rose from 2019 to 2024; after subtracting the market, insider buying fell from +0.028R to −0.079R per trade.
- **Luck did not replicate.** Earnings drift (+0.156R on 149 stocks) fell to +0.025R on 250 stocks it had never seen.
- **Costs were not the reason.** Several ideas showed no edge even with trading costs set to zero, and the largest tests had about 100,000 trades, so the negatives are precise.
- **A near miss is not a pass.** Short-sale volume (M16) was positive on both stock sets and in both halves of the sample and failed only because its interval touched zero; it is reported as not eligible.

What this means. For traders, a strategy shown online is one run, chosen after the fact; it should be held to the same tests before any money follows it. For anyone building backtests, the four safeguards that mattered most here are cheap to adopt: write the rules down first, count every try, subtract the market, and repeat the test on data that played no part in choosing the rule. For this project, the free-data, short-term route is closed; the method remains in place for the next question.

How to read this paper. Section 1 explains why Sift was built and how it works, Section 2 describes the seventeen ideas in plain words, Section 3 sets out the method, Sections 4 and 5 report and interpret the results, and Section 7 states what the study cannot show. A glossary of every abbreviation is in Appendix A.

1 Introduction

1.1 Motivation

Strategies spread online because they are easy to describe and easy to believe: a bot that doubled a small account overnight, a pair of indicators with a high win rate, a screenshot of a good week. What is missing is a fair test. Two traps make almost any idea look good:

- **Selection luck.** Try enough rules on one stretch of history and some will fit it by chance. This is called overfitting, or data-snooping, and the best-looking result of many tries is the one most likely to be luck.
- **The market's own rise.** In a period when stocks go up, almost any rule that buys will make money. That is not skill.

The goal was therefore twofold: build a pipeline that could find a short-term edge in US stocks, and make it hard for the pipeline to fool its builder. The rule was: **no real money until an idea has proved itself on data it was not tuned on.** Starting capital is $0, so everything uses free data and paper trading. The name, Sift, is the method: put each idea on a sieve and see what stays behind. So far, mostly chaff.

1.2 How Sift works

Sift has five parts, in the order data moves through them:

1. **Data.** A recorder downloads free market data (Alpaca's free tier gives consolidated-tape bars back to 2016), SEC filings, government contract records and public archives of posts, and stores them locally. Stock splits are applied when data is read, so a new split never corrupts history.
2. **Strategies.** Each idea is turned into exact rules that output trade signals: a direction, an entry, a stop-loss and a profit target, with the target fixed at 1.5 times the risk.
3. **Backtest engine.** It replays history trade by trade with realistic conventions: orders fill at the next bar's open plus slippage, a stop wins when a bar touches both stop and target, and gaps fill at the open. Results are measured in R, the profit as a multiple of the amount risked.
4. **Statistics.** Parameters are chosen on earlier years only and judged on later ones (walk-forward). Every configuration tried is logged, and a Deflated Sharpe check asks whether a result survives that many attempts. The market is subtracted, and promising results must repeat on stocks the rule never saw.
5. **Paper trading.** A loop that places orders on a broker's paper account was built and tested for plumbing, ready for the day an idea passes. None did, so nothing was ever run for evidence.

The work was organised as one **milestone per idea**. For each, a protocol (the exact rules, the parameter grid and the pass criteria) was written and committed to version control first; the test was then run once, and the result was recorded whatever it showed. The commit history therefore proves the order. The code, the analysis and the drafting were done with an AI coding assistant; the decisions, and every number in this report, trace back to results recorded in the repository.



Figure 1: The pipeline. Data flows from free sources through a split-safe store into strategies and a cost-aware backtest; only an idea that passes every check reaches the holdout, and only after that paper trading.

Terms. Alpaca = a US broker whose free tier supplies the price data. Bars = price candles (open, high, low, close, volume) for one time period. SEC = the US Securities and Exchange Commission; EDGAR = its public database of company filings. Split-safe = stock splits are adjusted when data is read. Stop = an order that exits a losing trade; target = an order that exits a winning trade. Backtest = replaying history to see what a rule would have done. Walk-forward = choosing settings on earlier years and testing them on later ones. Holdout = recent data kept unseen until the end. Paper trading = simulated trading on a broker's practice account. Trials counted = every configuration tried is logged.

1.3 Constraints fixed at the start

- US stocks, short-term horizons (minutes to a few weeks), a fixed 1:1.5 risk-to-reward bracket on trades.
- Start with $0 of capital. Free data only: Alpaca's free tier (consolidated-tape bars from 2016, 15 minutes delayed), SEC EDGAR filings, USAspending.gov, public archives of presidential posts.
- Universe chosen by a fixed rule (median dollar volume in Q1 2016), never by hand: 10 large caps, then 50, 150 and 250 further stocks (about 410 in total).

2 The seventeen ideas

Each idea is a belief about how prices behave. A test asks one question: if I had acted on that belief with fixed rules, would I have made money after costs, and was that more than just holding the market? The ideas came from four places: classic day-trading lore (ideas 1 to 4), academic research (5, 6, 11 and 12), indicator "bots" shown in online videos (9 and 10), and the intuition that politics moves stocks (7 and 8). Five more (13 to 17, milestones M14 to M18) came from published factor research and were tested after the first twelve, on new free data.

2.1 Short-term price patterns (minutes to hours, 5-minute bars)

1. **Opening-range breakout.** *The belief:* the first minutes of the trading day set a range, and when price closes beyond it, the move tends to continue. *The test:* trade in the direction of the break with a stop and a target at 1.5 times the risk, and close the trade by the end of the day. Optional filters were unusual volume, an overnight gap and the market's trend. Tried on 10 large stocks, then on 50 fresh ones.
2. **Failed-breakout fade.** *The belief:* a breakout that fails, with price closing back inside the range, traps the traders who bought it and reverses. *The test:* bet against the failed move, with the stop just beyond its extreme.

2.2 Daily swing trades (days to weeks, daily bars)

3. **RSI(2) mean reversion.** *The belief:* after a sharp two-day fall a stock tends to bounce within days, and to fall back after a sharp rise. *The test:* buy when the 2-day RSI, a gauge of very short-term momentum, is extremely low and sell short when it is extremely high, optionally only in the direction of the 200-day trend.
4. **20/200-day pullback.** *The belief:* in an established uptrend, a dip back to the 20-day average is a chance to buy cheaply, and the reverse in a downtrend. *The test:* trade the first close back to the 20-day average while the 20-day and 200-day averages agree on the trend.

2.3 Events and news (daily bars)

5. **Earnings drift.** *The belief:* when a company reports results and its stock jumps or falls by an unusual amount, it keeps moving the same way for days or weeks. *The test:* take the timing from the exact moment of the SEC filing of each earnings release, measure the market-adjusted reaction in multiples of the stock's normal daily range, and follow (or fade) reactions of one to three ranges. The drift is reported to have faded outside the smallest stocks [4].
6. **Insider buying.** *The belief:* when company officers and directors buy shares with their own money they may know something the market does not, especially if the purchase is out of character ("opportunistic") rather than a habit ("routine"). *The test:* trade clusters of purchases at the open after the public SEC filing, with the routine and opportunistic split from an academic paper by Cohen, Malloy and Pomorski [2].
7. **Presidential posts.** *The belief:* when the US president praises or attacks a listed company on social media, its stock moves, and a slower trader can still profit. *The test:* find posts that name a company (tweets 2016 to 2018, Truth Social 2022 to 2024), take their tone from fixed word lists, enter at the first open after the post, hold five sessions.
8. **Pentagon contract awards.** *The belief:* a large defence contract lifts the stock of the winner. *The test:* use awards of $100 million or more to seven listed contractors from USAspending.gov, buy at the open two sessions after the award date and hold five sessions.

2.4 Indicator bots popular in online videos (5-minute bars)

9. **UT Bot + Schaff Trend Cycle.** *The belief:* a trailing-stop indicator (UT Bot) flipping direction marks the start of a trend, and a second indicator (Schaff Trend Cycle) confirms it. *The test:* the common open-source definitions, 18 settings, including the video's asymmetric setup with different long and short parameters.
10. **Supertrend + Waddah Attar Explosion.** *The belief:* a trend line (Supertrend) agreeing with a momentum-surge indicator (Waddah Attar) marks high-quality entries, optionally only when the market is trending (ADX or Choppiness filter). *The test:* 12 settings, with the stop distance used in the video.

2.5 Volume (daily bars)

11. **Volume-dry-up reversal.** *The belief:* a stock that fell (or rose) against the market over a week on shrinking volume was pushed by impatient traders rather than news, and tends to snap back. *The test:* trade against such moves of two to three daily ranges, with and without the volume condition.
12. **High-volume continuation.** *The belief:* a stock whose volume is the highest in weeks is receiving new attention and tends to drift upward over the following weeks. *The test:* buy on the highest-volume day of the last 20 or 50 sessions, long only.

One more test is a side question, not an idea (Section 6): whether a 200-day trend filter makes simply holding the market safer.

What was expected. Where a protocol stated a chance of passing in advance, it was low: about 3 to 5% for each indicator bot, about 10% for insider buying, 12% and 8% for the two volume ideas, and 3% and 2% for the two political ones. The outcomes were consistent with those priors, and none of the tests was chosen because it was expected to win.

2.6 Five more ideas on new free data (milestones M14 to M18)

After the first twelve, Sift gained a forward data recorder and several new free sources (SEC filings, FINRA short-sale volume, XBRL share counts), and five more ideas were tested under the same rules. Four are monthly long-short baskets (buy the top fifth of stocks on a signal, short the bottom fifth, 5 basis points a side, trade at the first open of the month); one is an event study.

13. **8-K bad-news items (M14).** *The belief:* a negative company filing keeps pushing the stock down for days. *The test:* short on the next open after items such as defaults, delistings or restatements, hold five sessions.
14. **52-week-high momentum (M15).** *The belief:* stocks near their yearly high keep outperforming those far below it. *The test:* long the fifth nearest the high, short the fifth farthest, rebalanced monthly.
15. **FINRA short-sale volume (M16).** *The belief:* stocks with a high share of short selling underperform the next month. *The test:* rank by short-sale share of volume, long the lowest fifth, short the highest.
16. **Net share issuance (M17).** *The belief:* companies that issue shares underperform those that buy them back. *The test:* rank by one-year change in shares outstanding, long the lowest, short the highest.
17. **Lazy Prices (M18).** *The belief:* a company that rewrites the risk-factor section of its annual report is warning of trouble. *The test:* rank by how much Item 1A text changed against the prior year, long the least changed, short the most.

The first run of M15's replication exposed a data bug, described in Section 4.4.

3 Methodology

Most backtests that look good are noise that was selected. The framework was built to make that hard: Figure 2 shows the method applied to every idea, and Table 1 lists its safeguards.

Pass criteria (all must hold): at least 300 out-of-sample trades; the 95% interval of expectancy above zero; Deflated Sharpe at least 0.95; positive in at least 4 of 6 test years; still positive at double costs; at least 70% of neighbouring parameter settings positive; and, from M8 on, a market-adjusted interval above zero. A test with too few trades is "inconclusive", not a pass.

A note on units: **R** is profit measured in the amount risked on the trade. +0.1R means the average trade made 10% of what it risked, after costs.

3.1 Reading a result

A worked example shows why the 1.5 times target matters. Suppose each trade risks $100, the distance from the entry to the stop. A winner that reaches the target earns $150, which is +1.5R; a loser costs $100, which is −1R. The break-even win rate is therefore 40% (0.4 x 1.5 - 0.6 x 1 = 0). A rule can win 46% of its trades and still earn almost nothing: the insider-buying test won 45.9%, but its wins averaged +1.26R and its losses −1.02R, which nets to +0.03R before looking at the market.

A test's result is its average R per trade, the expectancy, with a 95% confidence interval: the range in which the true average very probably lies. "+0.028R ±0.078" means the true value could be anywhere from −0.050R to +0.106R, so it cannot be told apart from zero. A result counts as an edge only when the whole interval sits above zero.

Table 1: Safeguards built into every test.

Safeguard	What it does
Pre-registration	For each test the protocol, parameter grid and pass criteria were committed to version control before the run. The commit history shows the order, so it cannot be quietly changed afterwards.
Walk-forward, out of sample	Parameters are chosen on earlier years only and judged on the next. Test years were 2019 to 2024.
Untouched holdout	Data from 2025 onward is kept for a single final run, and only for an idea that has already passed everything. The code refuses to read it without an explicit flag. No analysis has read it.
Trial counting and Deflated Sharpe	Every configuration tried is logged (298 when the first twelve ideas were judged, 304 after M14 to M18). The Deflated Sharpe Ratio (Bailey and L'opez de Prado [1]) asks whether a result still stands after paying for the number of attempts.
Realistic fills and costs	Next-bar fills, stops assumed hit first when a bar is ambiguous, gaps filled at the open, per-share slippage on market fills, limit targets that must trade through to fill, stock splits handled at read time, one open trade per symbol. A stress test doubles costs.
Market-adjusted check	Added after earnings drift: each trade's result minus what holding SPY (the exchange-traded fund that tracks the S&P 500 index) the same way over the same dates would have earned.
Replication on unseen stocks	A single frozen setting is run once on stocks that took no part in choosing it.

Table 2: Walk-forward schedule: each test year is judged with a setting chosen on earlier years only.

Test year	Setting chosen using data from
2019	2016 to 2018
2020	2016 to 2019
2021	2016 to 2020
2022	2016 to 2021
2023	2016 to 2022
2024	2016 to 2023

3.2 How a trade is simulated

Each strategy emits a signal when a bar closes. The simulator then follows a fixed script, so that a rule cannot flatter itself:

1. **Entry.** A market order is filled at the next bar's open plus slippage, an assumed cost per share: one cent on liquid stocks, two cents on mid-sized ones, doubled in the stress test.
2. **Exit.** The stop and the target are set when the signal fires. If one bar touches both, the loss is counted, the cautious assumption. If price gaps past the stop, the exit is at the gap price, not at the stop. A profit target fills only when price trades through it, not when it merely touches it. Intraday trades are closed by 15:55 New York time; daily trades are closed at the close after a maximum holding period.
3. **Measurement.** Every simulated trade is scored in R, before any position sizing, so the statistics do not depend on how much money is assumed to be in the account. A separate portfolio view sizes each trade to risk 1% of the account under leverage and position caps.

3.3 How walk-forward works

Settings are never judged on the data that chose them. For each test year, the best-looking setting is picked using only earlier years and then run on the year it has not seen (Table 2):

The six test years are joined into one out-of-sample record, and that record is what the results below report. "Best-looking" means the highest average R minus one standard error, so a setting that did well on very few trades is penalised.

3.4 What the Deflated Sharpe check asks

The Sharpe ratio measures return per unit of risk. If hundreds of settings are tried, the best of them will have an impressive Sharpe ratio by luck alone, even if every one is worthless. The Deflated Sharpe Ratio estimates the probability that the best result is genuine after discounting for the number of attempts (298 at the time of the first twelve tests; 304 now); the criterion is at least 0.95. For a single frozen setting tested once on new stocks (M8) there is nothing to deflate for, so the check reduces to the probability that the true Sharpe ratio is above zero.

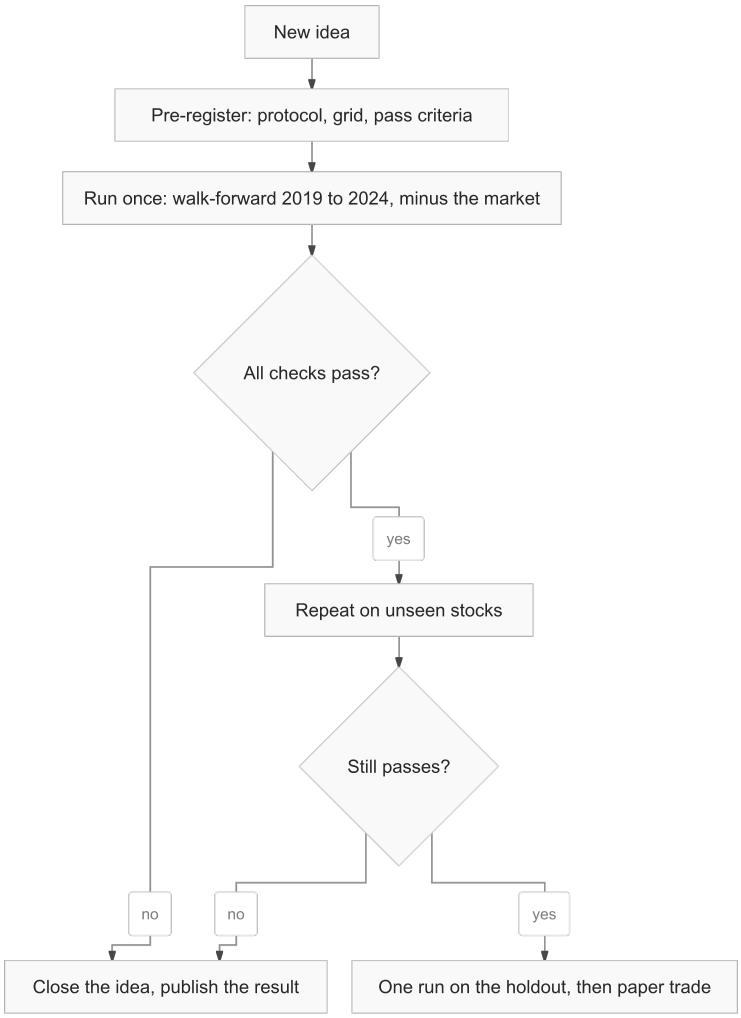


Figure 2: The method applied to every idea. Failing at any gate closes the idea and the result is published either way.

Terms. Pre-register = write down and commit the rules before running anything. Protocol = the exact rules and pass criteria; grid = the list of parameter settings to try. Walk-forward = settings chosen on earlier years, tested on later ones (Section 3). Minus the market = subtract what simply holding the market earned over the same dates. Every trial counted = each setting tried is logged and the final statistics account for the number of tries. Checks = the seven pass criteria listed in Section 3. Unseen stocks = stocks that took no part in choosing the setting. Holdout = data from 2025 onward, kept unread.

4 Results

Table 3 gives each idea's out-of-sample result and how many of its pre-registered checks it passed; Figure 3 shows the same results as a picture, with their 95% intervals.

M12 is an event study (average market-adjusted return over five sessions, net of 10 basis points), not a strategy sweep, so it has its own four criteria. "M4" and "M4b" are the ten-stock and fifty-stock runs of the same breakout grid.

Details, protocols and per-year numbers for each row are in `docs/m*-protocol.md` and `docs/m*-results.md`.

Table 3: Out-of-sample results of the first twelve tests, with the number of pre-registered checks passed. M14 to M18 are in Table 5.

Test	Idea	Data	Out-of-sample result (95% interval)	Checks passed
M4	Opening-range breakout	10 large caps, 5-min bars	+0.065R ±0.114	3 of 6
M4	Failed-breakout fade	same	−0.000R ±0.042	1 of 6
M4b	Opening-range breakout	50 fresh stocks	−0.046R ±0.028	1 of 6
M6	RSI(2) mean reversion	50 stocks, daily	−0.026R ±0.021	1 of 6
M6	20/200-day pullback	50 stocks, daily	−0.032R ±0.028	1 of 6
M7	Earnings drift (SEC filing times)	149 stocks, daily	+0.156R ±0.099	5 of 6
M8	M7's frozen setting, replicated	250 unseen stocks	+0.025R ±0.077	2 of 7
M9	UT Bot + Schaff Trend Cycle	50 stocks, 5-min	−0.038R ±0.006	1 of 6
M9	Supertrend + Waddah Attar	50 stocks, 5-min	−0.042R ±0.005	1 of 6
M10	Insider buying (SEC Form 4)	407 stocks, daily	+0.028R ±0.078	3 of 7
M11	Volume-dry-up reversal	409 stocks, daily	−0.008R ±0.010	1 of 7
M11	High-volume continuation	409 stocks, daily	+0.100R ±0.019	5 of 7
M12	Presidential posts naming a listed company	133 events	−0.05% ±0.60 per event	1 of 4
M12	Large Pentagon awards to listed primes	933 events	−0.10% ±0.27 per event	1 of 4

4.1 Reading the table

Each row is one test, run once. The result column is the average out-of-sample R per trade with its 95% interval, and the last column counts the pre-registered checks passed (six, or seven from M8 on). Three patterns stand out:

- **Intervals shrink with the number of trades.** The two indicator bots (M9) had about 100,000 trades each and intervals of ±0.005R, so their small negative results are firm. The ten-stock breakout (M4) had 435 trades and an interval of ±0.114R, wide enough to hide almost anything.
- **Positive is not the same as significant.** Only two ideas had an interval entirely above zero before the market was subtracted: earnings drift (M7) and high-volume continuation (M11).
- **Most failures were not close calls.** Seven of the first twelve ideas passed only the minimum-trade-count check.

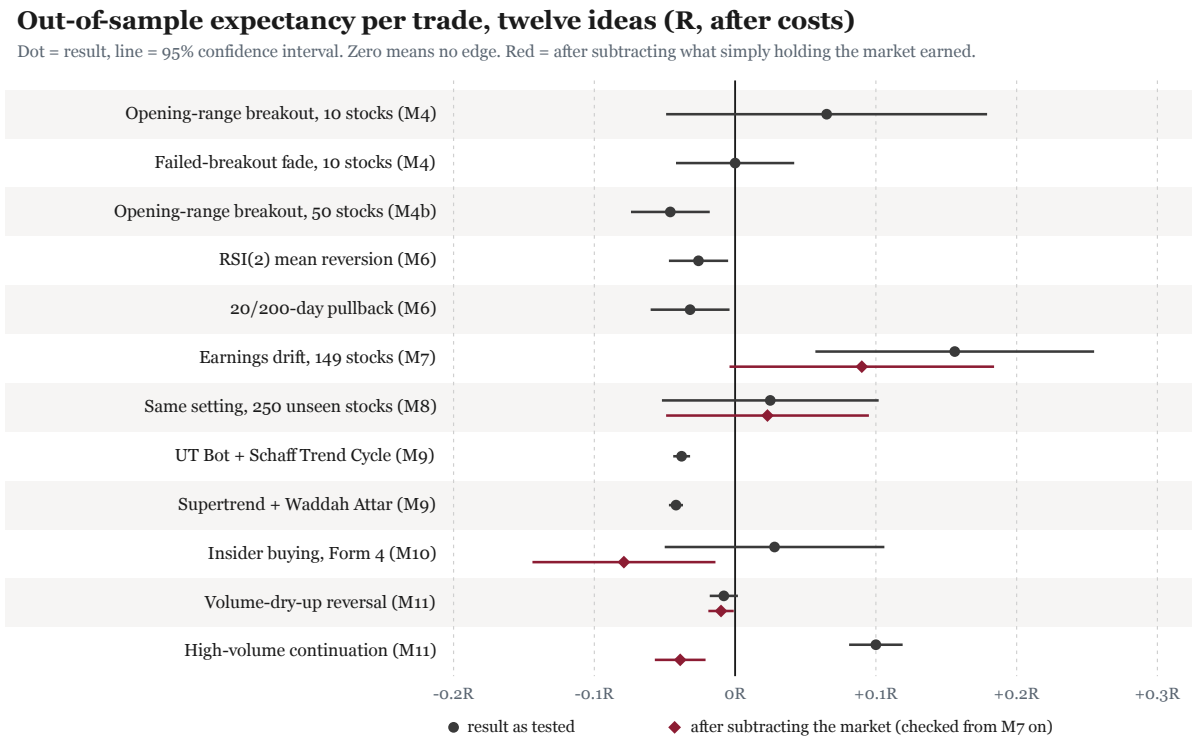


Figure 3: Out-of-sample result per trade, in R (profit measured in units of the amount risked). Zero means no edge. Grey dots: as tested. Red diamonds: after subtracting what simply holding the market earned over the same dates, a check added from M7 on. Lines are 95% confidence intervals.

Terms. M4 to M11 = the milestone numbers of the tests (see Table 3). RSI(2) = Relative Strength Index over 2 days, a gauge of very short-term momentum. 20/200-day = the 20-day and 200-day moving averages of the price. UT Bot = a trailing-stop trend indicator popular on charting sites; Schaff Trend Cycle (STC) = a momentum-cycle indicator. Supertrend = a trend line built from the average daily price range; Waddah Attar = a momentum-surge indicator. Form 4 = the SEC filing in which company insiders report their trades. Earnings drift = a stock keeps moving after its earnings reaction.

4.2 Which safeguard stopped each idea

Table 4 shows, for the first twelve strategy tests, which pre-registered check each idea passed (yes) or failed (no); the columns follow the seven checks of Section 3. A dash means the check did not exist yet: the market-adjusted comparison was added after earnings drift (M7).

Two checks did most of the work. Every idea had enough trades, but none passed the Deflated Sharpe check, and none of the four ideas tested against the market passed that comparison. The interval check alone would have let two ideas through; the market-adjusted comparison removed the rest of their gain. The M12 event studies and the M13 trend filter have their own criteria and are not in the matrix.

4.3 The two event studies (M12)

M12 asks a simpler question than the strategy tests: after the news, is any average move left to trade? For each event the study enters at the first open at which the news could be traded (a post before 09:30 New York time at that day's open, any other post at the next open, and a contract award at the open two sessions after its date, to err on the side of caution). It holds five sessions, subtracts what SPY did over the same dates and 10 basis points of cost, and flips the sign for a negative post. Four criteria were fixed in advance: at least 100 events; a positive average whose 97.5% interval lies above zero; a positive average in two thirds of the years with at least eight events; and a positive average in both halves of the sample.

Posts failed all but the first (average −0.05%, positive in only 2 of 6 years); awards also

Table 4: Which pre-registered check each idea passed (yes) or failed (no); a dash means the check did not exist yet.

Idea	Enough trades	Above zero	Deflated Sharpe	Years positive	2x costs	Neigh-bours	Market-adjusted
Opening-range breakout (M4)	yes	no	no	yes	yes	no	–
Failed-breakout fade (M4)	yes	no	no	no	no	no	–
Opening-range breakout, 50 stocks (M4b)	yes	no	no	no	no	no	–
RSI(2) mean reversion (M6)	yes	no	no	no	no	no	–
20/200-day pullback (M6)	yes	no	no	no	no	no	–
Earnings drift (M7)	yes	yes	no	yes	yes	yes	–
Earnings drift, replicated (M8)	yes	no	no	no	yes	no	no
UT Bot + Schaff Trend Cycle (M9)	yes	no	no	no	no	no	–
Supertrend + Waddah Attar (M9)	yes	no	no	no	no	no	–
Insider buying (M10)	yes	no	no	no	yes	yes	no
Volume-dry-up reversal (M11)	yes	no	no	no	no	no	no
High-volume continuation (M11)	yes	yes	no	yes	yes	yes	no

failed all but the first (average −0.10%, positive in 5 of 9 years). Splitting the posts by tone changed little: positive posts averaged −0.01% and negative posts −0.12%, where a negative number means the trade direction was wrong.

4.4 Five more tests (M14 to M18)

Table 5 gives the five later tests. They use the monthly-basket engine, so the unit is a percentage per month (or per event for M14), not R, and the criteria are the ones written into each protocol: enough observations, an interval above zero, positive in two thirds of years and in both halves, and replication on unseen stocks.

Test	Idea	Result	Verdict
M14	8-K bad-news items (879 events)	+0.09% ±0.32% per event	Not eligible
M15	52-week-high momentum	−0.37% a month	Not eligible
M16	FINRA short-sale volume	+0.38% (primary), +0.34% (replication) a month	Not eligible: interval spans zero
M17	Net share issuance	−0.22% (primary), +0.18% (replication) a month	Not eligible
M18	Lazy Prices (10-K risk-factor text)	+0.25% / +0.12% a month, 45 months in the book	Inconclusive

Table 5: Five further pre-registered tests, each run once. All used the original stock universes (ranks 1 to 150 and 151 to 400 by 2016 dollar volume).

Three things are worth recording:

- **A near miss is not a pass.** Short-sale volume (M16) was positive on both universes and in both halves and failed only because its interval touched zero. Sixty-four months cannot tell it from luck. It could justify a longer, separately pre-registered test; it does not justify re-reading this one.
- **A data bug can fake a result.** M15's replication first read −4.4% a month because four stocks had reverse splits missing from the split list and looked like +12,000% days. The fix (drop stocks with a one-session gain above +100%) came after seeing the result, and is disclosed as a deviation with both runs shown.
- **Too little data is its own verdict.** Lazy Prices populated only 45 of the 48 required months because 10-Ks cluster in February and March, so it is reported as inconclusive, not as a failure or a pass.

5 Discussion: what kept recurring

Costs were not the explanation. The first guess was that a one-cent spread against tight five-minute stops ate a small edge. A zero-cost diagnostic showed otherwise: breakouts were −0.020R ±0.028 before costs and volume reversal +0.006R ±0.010. There was no gross edge to eat, so cheaper execution would not have rescued these ideas.

The market did the work. The long-only families look strongest in 2019 to 2024, a period in which US stocks rose. The market-adjusted check changed the picture (Table 6):

All 16 volume-spike configurations were positive as tested, and they still failed. A related effect: the stock universe is made of companies that were liquid in 2016 and are still listed today. Survivors make long positions look better and short positions worse (in M6 longs were near zero and shorts clearly negative).

Lucky results did not replicate. The ten-stock breakout test (+0.065R) turned into −0.046R on fifty fresh stocks, with all 54 configurations negative. Earnings drift passed five

Table 6: The three strongest long-side results, before and after subtracting the market.

Idea	As tested	After subtracting the market
Earnings drift (M7)	+0.156R	+0.090R ±0.094
Insider buying (M10)	+0.028R	−0.079R ±0.065
High-volume continuation (M11)	+0.100R	−0.039R ±0.018



Figure 4: Why each idea failed, by main reason. Several had more than one: the breakout, for example, also showed a faint edge on 10 stocks that vanished on 50.

Terms. Before costs = the test was repeated with trading costs set to zero and still showed no gain. RSI 2 = 2-day Relative Strength Index. UT Bot = a trailing-stop trend indicator; STC = Schaff Trend Cycle, a momentum-cycle indicator. Supertrend = a trend line built from the average daily price range; Waddah = the Waddah Attar Explosion momentum indicator. Pullback = a dip back to a moving average inside a trend. Rising market = the gain disappeared once the market's own return was subtracted. New stocks = stocks that took no part in choosing the setting.

of six checks and then fell to +0.025R on 250 unseen stocks. The Deflated Sharpe check had flagged the first of these in advance; replication confirmed it.

Several attractive stories went nowhere. Distinguishing routine from opportunistic insider trades follows a well-known paper [2]; in our large and mid-cap universe there were too few opportunistic buys to test the split. Presidential posts: published work finds same-day reactions to tweets about firms [3], but starting at the next open, as a slower trader must, nothing was left.

5.1 A closer look: the near miss

Earnings drift came closest, so it is worth following step by step. On 149 liquid stocks it passed five of six checks: 551 trades, +0.156R ±0.099, positive in all six years and still positive at double costs. Only the Deflated Sharpe check failed (0.006), because the winning setting was one of 36 tried. About half of the raw gain was the rising market (+0.090R after subtracting it), and the drift appeared only from 2019 on. The test that settles such a case is replication: the same setting, frozen, on 250 stocks that had played no part in choosing it. There the average fell to +0.025R ±0.077 (+0.023R after the market), the year-by-year results turned into noise (+0.06, −0.06, −0.07, +0.12, +0.20 and −0.07R), and the measured drift in the price reaction itself shrank from +1.13% to +0.06%.

6 A side question: does a trend filter make holding the market safer?

Since the market kept beating the rules, I tested the simplest allocation rule: hold SPY while it is above its 200-day average, otherwise hold cash. One parameter, fixed in advance, 2017 to 2024 (Table 7):

Table 7: Holding SPY with and without the 200-day trend filter, 2017 to 2024.

	Buy and hold	200-day filter
Return per year	12.8%	8.8%
Worst drawdown	−32.4%	−21.8%
Sharpe (risk-adjusted, no cash yield)	0.78	0.75

It reduced the worst falls (2020: −19% instead of −32%; 2022: −17% instead of −27%) but missed the rebounds, and 18 of its 26 in-and-out trades lost money. It passed two of three pre-registered checks and failed the return one by about 4 points a year. Eight years and two crashes are a short sample, and this is research, not advice: how large a fall a holder can sit through matters more than these figures.

7 Limitations

Absence of evidence in these tests is not proof that no edge exists. The scope was narrow:

- **Free data and daily or five-minute bars.** I could not test the first seconds and minutes after news, where fast traders operate, nor order-book or options data, nor paid datasets (earnings estimates, intraday news).
- **Large and mid caps only.** Several published effects are strongest in small stocks, which were not tested. A survivorship-free universe that includes delisted tickers has since been built (the free data source does serve their bars), but none of the seventeen tests has yet been rerun on it.

- **One regime.** 2016 to 2024 is one stretch of one market, dominated by a rise. The holdout (2025 onward) was kept unread for a final run that no idea earned.
- **Survivor and hindsight bias.** The universes are stocks listed today. This was disclosed in each protocol and is why the market-adjusted check matters.
- **Clustering.** Events bunch in market selloffs, so confidence intervals are somewhat optimistic about independence.
- **Backtests are not live trading.** Slippage in fast markets, rejected orders and partial fills are modelled approximately. A paper-trading loop was built and tested against a broker's paper account for plumbing only; nothing was run for evidence.

8 Conclusion

- **No idea passed.** Of seventeen, the closest were earnings drift (5 of 6 checks, then it failed to replicate on 250 unseen stocks), a volume-spike rule (5 of 7 checks, then it turned negative after subtracting the market) and short-sale volume (positive on both universes, but with an interval that spans zero).
- **When it looked like an edge, it was mostly the market.** Long-only rules were positive in 2019 to 2024 largely because stocks rose. Insider buying went from +0.028R to −0.079R, and volume spikes from +0.100R to −0.039R, once the market was subtracted.
- **Costs were not the main reason.** The price-pattern families had no edge even at zero cost (for example breakouts at −0.020R, volume reversal at +0.006R, both consistent with zero).
- **Replication was the strongest filter.** A faint breakout result on 10 stocks (+0.065R) became −0.046R on 50 fresh ones. Earnings drift (+0.156R) became +0.025R on 250 unseen stocks.
- **Precision was high, so the negatives are real.** Some tests had around 100,000 trades with intervals of about ±0.005R. These are measured zeros and small negatives, not "not enough data".
- **A side question gave a trade-off, not a free lunch.** A 200-day trend filter on SPY cut the worst drawdown from −32% to −22% but cost about 4 points a year of return.

9 Recommendations

1. **Write the rules down before you look.** Commit them. It removes the most common way to fool yourself.
2. **Count your tries.** The best of many looks good by luck; deflate for it.
3. **Subtract the market.** In a rising period a long-only rule needs to beat holding, not zero.
4. **Replicate on stocks that took no part in choosing the rule.** It was the strongest filter here.
5. **Ask whether there is a gross edge before blaming costs.**
6. **Treat screenshots, reels and "I made X overnight" posts as hypotheses.** They are one run, picked after the fact, and rarely come with losing runs or costs. They deserve the same test as anything else, and most will fail it.
7. **A clear "no" is a result.** It saved capital, and the method now makes the next question cheap to test.

10 Reproducibility

Everything is in the project repository: code, each protocol (committed before its run), each result write-up and the tests (872 passing). Data is not redistributed; the scripts fetch it from the free sources above.

```
uv sync
uv run pytest
uv run python scripts/sweep.py --config config/sweeps/m10-insider.toml
```

The last command reruns one pre-registered sweep (insider buying).

Start with `README.md`, then `docs/m4-protocol.md` for the method, and `docs/m*-results.md` for each result.

A Glossary

Every abbreviation and technical term used in the paper, in alphabetical order.

ADX	Average Directional Index: a measure of how strongly a market is trending.
ATR	Average True Range: the typical daily price range of a stock, used to size stops.
Backtest	Replaying historical data to see what a fixed rule would have done.
Bar	A price candle for one period: open, high, low, close and volume.
Basis point (bp)	One hundredth of one percent (0.01%).
Choppiness Index	A measure of whether a market is ranging rather than trending.
Confidence interval	The range in which the true average very probably lies; here 95% unless stated.
Deflated Sharpe Ratio	The probability that the best Sharpe ratio is genuine after accounting for the number of tries.
EDGAR	The SEC's public database of company filings.
Event study	Measuring the average price move after a type of event, such as an earnings release.
Expectancy	The average result per trade, here in R.
Form 4 / 8-K	SEC filings: insider trades (Form 4) and material company events such as earnings (8-K, Item 2.02).
Holdout	Recent data (2025 onward) kept unread for a single final test.
Out of sample	Data that played no part in choosing a rule's settings.
Paper trading	Simulated trading on a broker's practice account, with no real money.
Pre-registration	Writing down and committing the rules and pass criteria before running a test.
R	Profit measured as a multiple of the amount risked on the trade.
RSI	Relative Strength Index: a gauge of recent momentum; RSI(2) uses two days.
SEC	The US Securities and Exchange Commission.
Sharpe ratio	Average return divided by its volatility: return per unit of risk.
Slippage	The cost of trading at a worse price than the last quote, assumed per share.
SPY	An exchange-traded fund that tracks the S&P 500 index; used as "the market".
STC	Schaff Trend Cycle: a momentum-cycle indicator.
Supertrend	A trend line built from the average true range.

Survivorship bias	Distortion from studying only companies that still exist today.
UT Bot	A trailing-stop trend indicator popular on charting sites.
Waddah Attar Explosion	A momentum-surge indicator with a dead zone based on ATR.
Walk-forward	Choosing settings on earlier years and judging them on the next, year by year.

B Index of protocols and results

Each test has a protocol, committed before the run, and a result write-up, in the `docs` folder of the repository.

Table 8: Where to find each test's protocol and result (Markdown files in the `docs` folder).

Test	Idea	Protocol	Result
M4	Opening-range breakout, failed-breakout fade	`m4-protocol`	`m4-results`
M4b	Breakout on 50 fresh stocks	`m4b-protocol`	`m4b-results`
M6	RSI(2) reversion, 20/200-day pullback	`m6-swing-protocol`	`m6-results`
M7	Earnings-announcement drift	`m7-events-protocol`	`m7-results`
M8	The same setting on 250 unseen stocks	`m8-confirmation-protocol`	`m8-results`
M9	UT Bot + STC, Supertrend + Waddah Attar	`m9-indicators-protocol`	`m9-results`
M10	Insider buying	`m10-insider-protocol`	`m10-results`
M11	Volume reversal and continuation	`m11-volume-protocol`	`m11-volume-results`
M12	Presidential posts, Pentagon awards	`m12-political-protocol`	`m12-political-results`
M13	200-day trend filter on SPY	`m13-trend-filter-protocol`	`m13-trend-filter-results`
M14	8-K bad-news items	`m14-eightk-protocol`	`m14-eightk-results`
M15	52-week-high momentum	`m15-high52-protocol`	`m15-high52-results`
M16	FINRA short-sale volume	`m16-shortvol-protocol`	`m16-shortvol-results`
M17	Net share issuance	`m17-issuance-protocol`	`m17-issuance-results`
M18	Lazy Prices (10-K risk-factor text)	`m18-lazyprices-protocol`	`m18-lazyprices-results`

M0 to M3 and M5 are infrastructure: decisions, a data-depth probe, the recorder, the backtest engine and the paper-trading loop.

Disclosures

This is a personal research project. It is not investment advice and recommends no security or strategy. No real money was traded. The code, analysis and drafting were done with an AI coding assistant (Claude Code); every number in this paper is taken from results recorded in the repository, and the protocols' commit order is the evidence of what was decided when.

References

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[3] C. Ajjoub, T. Walker and Y. Zhao, "Social media posts and stock returns: The Trump factor," *International Journal of Managerial Finance*, 17(2), 185–213, 2021. https://econpapers.repec.org/RePEc:eme:ijmfpp:ijmf-02-2020-0068

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